

Project Initialization and Planning Phase

Date	1 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

The **OptiCrop – Smart Agricultural Production Engine** project aims to understand the challenges faced by farmers while selecting the most suitable crop for cultivation. This customer problem statement helps identify the needs, pain points, and expectations of farmers so that an AI-based crop recommendation system can provide accurate and useful suggestions. The system analyzes soil nutrients and environmental conditions to recommend the best crop, helping farmers improve productivity and reduce the risk of crop failure.

OptiCrop – Smart Agricultural Production Engine

The OptiCrop project focuses on understanding the challenging need by farmers while selecting suitable crops. This includes problems such as unpredictable weather, lack of soil nutrients, pest attacks, improper irrigation, and delays in AI-based crop recommendation. OptiCrop aims to provide smart solutions by integrating data AI-based tool. Real-time environmental data, analysis of soil conditions, and recommendations are presented as outputs by understanding the farming problem. The system provides accurate crop recommendations based on factors like soil type, water availability, and seasonal patterns.

Customer Problem Statement Template

Which makes me feel	Because	But	I'm trying to	I am (Customer)	Problem Statement (PS)
 Worried about uncertain weather and its impact on my farm yield	 Soil conditions and weather uncertainty make it hard to decide the best crop to grow this season	 I want a smart solution that can guide me based on my soil, land, and weather	 Select the best crops to improve my yield	 A farmer who cultivates different crops	PS-1
 Frustrated because my traditional farming methods are not working well	 I lack knowledge about best crop practices and modern farming techniques	 I need a solution that provides easy and quick recommendations	 Increase productivity and earn better income	 Someone who wants to improve my farming results	PS-2
 Confused by too much information available from different sources	 It is not easy to understand which crop is right for my soil and season	 I want an AI-based tool that gives clear and accurate advice in simple language	 Make smart decisions and save time and resources	 A tech-savvy farmer who uses digital tools	PS-3

Example:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	Select the most suitable crop for my farmland	I don't know which crop is suitable for my soil and	Soil nutrients and climate vary from	Confused and worried about crop failure

			weather conditions	season to season	
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